Beijing-Dublin International College

SEMESTER I FINAL EXAMINATION - 2016/2017

Beijing-Dublin International College

《UNIVERSITY PHYSICS 1》 BDIC1015J

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Time Allowed: 95 minutes

Instructions for Candidates

Full marks will be awarded for complete answer to All questions.

BJUT Student ID: ________________ **UCD Student ID: ________________**

I have read and clearly understand the Examination Rules of both Beijing University of Technology and University College Dublin. I am aware of the Punishment for Violating the Rules of Beijing University of Technology and/or University College Dublin. I hereby promise to abide by the relevant rules and regulations by not giving or receiving any help during the exam. If caught violating the rules, I accept the punishment thereof.

Honesty Pledge：____ (Signature)

Instructions for Invigilators

Non-programmable calculators are permitted.

No rough-work paper is to be provided for candidates.

Obtained score

Part 1: Multiple Choice (2 marks each question, 30 marks altogether)

1. A mass point executes curvilinear movement on a plane. Its instantaneous velocity is $\vec{v}$, instantaneous speed is v, average speed is $\bar{v}$, and average velocity is $\bar{\vec{v}}$. Which of the following relationships is right?

(A) $|\vec{v}| = v, |\bar{\vec{v}}| \neq \bar{v}$ (B) $|\vec{v}| \neq v, |\bar{\vec{v}}| \neq \bar{v}$

(C) $|\vec{v}| = v, |\bar{\vec{v}}| = \bar{v}$ (D) $|\vec{v}| \neq v, |\bar{\vec{v}}| = \bar{v}$

2. There are two particles A and B, $2m_A = m_B$. At $t = 0$ the velocity of particle A is $\vec{v}_{A1} = 3\vec{i} + 4\vec{j}$ (m/s) and the velocity of particle B is $\vec{v}_{B1} = 2\vec{i} - 7\vec{j}$ (m/s). Assume there is no external force acting on the two-particle system. Because the interaction between the two particles the velocity of particle A becomes $\vec{v}_{A2} = 5\vec{i} - 4\vec{j}$ (m/s), then the velocity of particle B is

(A) $\vec{v}_{B2} = 2\vec{i} - 7\vec{j}$ (m/s).

(B) 0

(C) $\vec{v}_{B2} = \vec{i} + 3\vec{j}$ (m/s).

(D) $\vec{v}_{B2} = \vec{i} - 3\vec{j}$ (m/s) .

3. Identical constant forces push two masses, A and B to the right, continuously over identical surfaces for identical periods of time. A is initially at rest and B is initially moving to the right. If the two masses are identical, which one undergoes a larger kinetic energy change?

(A) A has a larger kinetic energy change than B.

(B) A and B have equal kinetic energy changes.

(C) B has a larger kinetic energy change than A.

4. The change in mechanical energy of a closed system is equal to the work done on it by

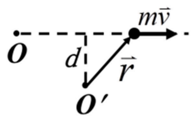
(A) all forces acting in the system.

(B) all conservative forces acting in the system.

(C) all non-conservative forces acting in the system.

(D) all the external forces and the internal non-conservative forces acting in the system.

5. A particle with mass m moves with constant velocity $\vec{v}$. The magnitude of the angular momentums of this particle with respect to O and O' are



(A) 0 and 0.

(B) 0 and mvd .

(C) mvd and 0.

(D) mvd and mvd .

6. A particle on a spring executes simple harmonic motion. If the mass of the particle and the amplitude are both doubled then the period of oscillation will change by a factor of

(A) 4 (B) $\sqrt{8}$ (C) 2 (D) $\sqrt{2}$

7. The equation of a simple harmonic wave is $y = A\cos(at - bx)$, where a and b are positive constants. Then

（A）the frequency of the wave is a （B）the period of the wave is $\frac{2\pi}{a}$

（C）the wavelength of the wave is $\frac{\pi}{b}$ （D）the wave speed is $\frac{b}{a}$

8. Two events *A* and *B* are simultaneous but occur at different locations in frame *S*. In any other frame S' moving relative to frame *S* at a constant velocity in the direction of positive x,

(A) the events could be simultaneous and occur at the same location.

(B) the events could either be simultaneous or occur at the same location, but not both.

(C) the events could be both simultaneous and at the same location.

(D) the events will be neither simultaneous nor at the same location.

9. Three containers contain same ideal gases with same molecular number density n. If the ratio of rms speed $\sqrt{\overline{v_1^2}} : \sqrt{\overline{v_2^2}} : \sqrt{\overline{v_3^2}} = 1:2:4$, then the ratio of the pressure $p_1 : p_2 : p_3$ is

(A) $1:2:4$ (B) $4:2:1$

(C) $1:4:16$ (D) $1:4:8$

10. A thermally insulated box is divided by a partition into two compartments, each having volume V. Initially, one compartment is filled with an ideal gas at temperature *T*, and the other compartment is evacuated. Then we break the partition, and the gas expands to fill both compartments. Which statement is right to describe the changes of temperature *T* and entropy *S* in this free adiabatic expansion process?

(A) Both the temperature *T* and entropy *S* decrease.

(B) The temperature *T* rises and the entropy *S* increases.

(C) The temperature *T* rises and the entropy *S* remains the same.

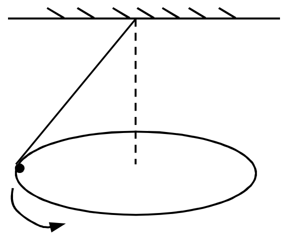
(D) The temperature *T* remains the same and the entropy *S* increases.

Obtained score

Part2: Blank Filling (5 marks each question, 20 marks altogether)

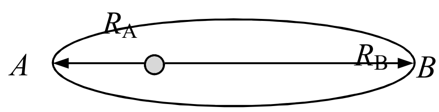
11. (4 marks)A particle moves in a circle with radius $R = 2\text{m}$. The tangential acceleration of the particle is $a_t = 2\text{m/s}^2$. The centripetal acceleration of the particle when $t = 1\text{s}$ is__________. The distance the particle goes through from $t = 0$ to $t = 1\text{s}$ is___________.

12. (6 marks) As shown in the figure is a conical pendulum. The ball with mass m rotates in horizontal plane with a constant angular velocity ω. In the process of one complete revolution, the magnitude of the momentum increment of the ball is___________; the magnitude of the impulse of the gravity of the ball is____________; and the magnitude of the impulse of the tension in the rope is_______________.



13. (3 marks) Suppose a particle takes part in three simple harmonic motions of the same frequency in the same direction respectively expressed by $x_1 = A\cos(\omega t + \pi/3)$, $x_2 = A\cos(\omega t + 5\pi/3)$ and $x_3 = A\cos(\omega t + \pi)$, the equation of the resultant motion is____________.

14. (3 marks) An artificial satellite *m* carries out an elliptical motion around the Earth. Point *A* is at perihelion and point *B* is at aphelion. The distances from points *A* and *B* to the center of the Earth is r_1 and r_2. The difference of kinetic energy E_{kB}-E_{kA} is_______________ (*M* is the mass of the Earth, *G* is gravitational constant)



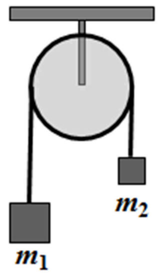
15. (4 marks) A quantity of gas containing N molecules has a speed distribution function $f(v)$. The number of molecules with speeds between v_1 and v_2 at an equilibrium state is __________. The average speed $\bar{v}$ at equilibrium state is ____________.

Obtained score

Part 3: Calculation(10 marks each question, 50 marks altogether)

16. A bullet with mass m is projected out horizontally into the sand with speed v. Assume it is subject to the resistance $f = -kv$ （k is a constant）. Ignore the gravity of the bullet. Find (a) the function of the v verses x of the bullet after it enters into the sand; (b) the maximum depth that the bullet can reach.

17. A pulley can be considered as a uniform disk of mass M and radius R, mounted on a fixed (frictionless) horizontal axle. Two blocks of mass m_1 and m_2 ($m_1 > m_2$) hang from a light cord that crosses the rim of the pulley (Suppose the cord does not slip or stretch). If the system is at rest in the beginning, calculate the angular speed of the pulley at time *t*. （The moment of inertia of the pulley is $J = \frac{1}{2}MR^2$）



18. A mass point executes simple harmonic motion in x direction around the equilibrium position x=0 with frequency of 0.25Hz. The speed of point x=0 is v=+1.57cm/s when t=0. Find the equation of simple harmonic motion.

19. The linear density of a rod (length L and mass m) is ρ. (a) Suppose the rod moves along the length direction with a speed of v, what is the linear density of the rod? (b)If the rod moves perpendicular to the length direction with a speed of v, what is the linear density of the rod?

20. As shown in the figure, a certain amount of ideal gas undergoes a cyclic process. If T_A=300K, (a) find T_B and T_C; (b) what are the works done by the gas in these three process? (c) Find the total heat energy absorbed in a complete cycle.